**A** XYZW **B** YWZX **C** XZWY **D** YXWZ

**11** A student sets up a simple harmonic oscillator comprising a mass *m* and a light spring with force constant *k* which oscillates with a period *T*. At equilibrum, the extension of this spring is $x_0$.

To overcome air resistance, the length of the spring is halved and the oscillator is attached to an actuator (driver). At the new equilibrium point, the extension of this spring is now $\frac{1}{2}x_0$.

What frequency should the actuator be set so that the oscillator can achieve maximum amplitude?

**A** $\frac{1}{T\sqrt{2}}$ **B** $\frac{\sqrt{2}}{T}$ **C** $\frac{1}{T}$ **D** $\frac{2}{T}$